Practical No: 02

Title:

Enterprise Architecture with emerging technologies such as ***Cloud/IOT/AI/Blockchain***.

Problem Statement:

Sketch enterprise architecture with emerging technologies such as cloud / IoT /AI / Block chain.

Software Tools: Archi (ArchiMate), Draw.io, EDrawMax; optional: PlantUML for sequence diagrams.

Theory:

EA aims to align business objectives with technology capabilities to achieve strategic goals. EA provides a holistic view of an organization's structure, processes, information systems, and technology infrastructure. The primary function of EA is to ensure that all the components of the enterprise, including business strategies, business processes, data architectures, and system architectures, are integrated, secure, and efficient.

Modern Enterprise Architecture (EA) integrates emerging technologies such as Cloud Computing, Internet of Things (IoT), Artificial Intelligence (AI), and Blockchain to improve business processes, increase efficiency, and support digital transformation. These technologies help organizations build scalable, secure, and intelligent enterprise systems.

1. Cloud Computing in Enterprise Architecture

Cloud Computing is the delivery of computing services (servers, storage, databases, networking, software, and analytics) over the internet on a pay-as-you-go basis.

Role in Enterprise Architecture

Cloud technology enables enterprises to build flexible and scalable IT infrastructures while reducing hardware and maintenance costs.

Example

A bank hosts its internet banking system on a cloud platform to provide uninterrupted services to customers worldwide.

2. Internet of Things (IoT) in Enterprise Architecture

IoT is a network of physical devices embedded with sensors, software, and communication technologies that collect and exchange data.

Role in Enterprise Architecture

IoT connects physical assets with enterprise systems, enabling real-time monitoring and automation.

Example

A manufacturing company installs IoT sensors on machines to monitor temperature and vibration, reducing equipment failures.

3. Artificial Intelligence (AI) in Enterprise Architecture

Artificial Intelligence refers to computer systems that simulate human intelligence to perform tasks such as learning, reasoning, prediction, and decision-making.

Role in Enterprise Architecture

AI enables intelligent automation, predictive analytics, and data-driven decision-making across enterprise systems.

Example

An e-commerce company uses AI to recommend products based on customer purchase history.

4. Blockchain in Enterprise Architecture

Blockchain is a decentralized and distributed digital ledger that securely records transactions across multiple computers.

Role in Enterprise Architecture

Blockchain provides transparency, security, trust, and traceability in enterprise business processes.

Example

A pharmaceutical company uses blockchain to track medicines from manufacturers to hospitals, ensuring authenticity.

Characteristics of Blockchain

1. **Decentralization**
    - Data is stored across multiple nodes instead of a single central server.
    - Eliminates the single point of failure.
2. **Transparency**
    - All authorized participants can view transactions.

- Increases accountability.

3. **Immutability**
   - Once a transaction is added, it cannot be modified or deleted.
   - Ensures data integrity.
4. **Security**
   - Uses cryptographic algorithms to protect data.
   - Transactions are digitally signed and encrypted.
5. **Consensus Mechanism**
   - Transactions are validated by network participants before being added.
   - Prevents fraudulent transactions.
6. **Traceability**
   - Every transaction has a complete history.
   - Useful for auditing and supply chain management.

Example:

Cloud-based solution

A **cloud-based solution** is an integrated set of IoT devices, components, and services, that addresses a business need and that connects devices directly to the cloud. An example of a cloud-based solution is a fleet of delivery trucks that send sensor data to the cloud for analysis and visualization:

